

Fig. 4: Actual-size PCB pattern of the vibration-activated smart CRO probe

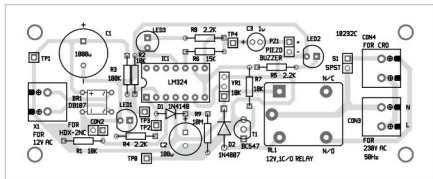


Fig. 5: Component layout of the PCB

(pin 5 of A2) decreases beyond the set voltage (at pin 6 of A2), output of A2 at its pin 7 goes low. Output of A2 is connected to relay driver transistor T1.

When A2 output is high, relay RL1 energises and the CRO gets AC mains power supply via relay contacts to operate it. At the same time, the CRO-on indicator LED2 glows.

On time of the CRO can be set as per requirement using VR1. With a small voltage set via VR1 at inverting terminal pin 6 of A2, greater will be the delay. Similarly, with more voltage set, smaller will be the delay.

An extra op-amp (A3) is used to provide acknowledgment to the user that the CRO has switched off after some idle time. For this, A3 is used in inverting-buffer mode. When output of A2 is high, output of A3 will be low, and vice versa.

When the CRO is off, output of A3 will be high. Due to this, the buzzer gets activated and produces a beep sound whose loudness gradually decreases within a short time. This

happens because C3 initially acts as a short circuit and charges to +12V and blocks further current passing through the buzzer. Increasing C3 value makes the buzzer beep longer. This acknowledgement is like a communication between the user and the circuit. LED3 also flashes when the beeping sound is produced.

Auto and manual modes are provided by mode switch S1. When S1 is open, the circuit is in auto mode. With S1 closed, the circuit is in manual mode and bypasses the relay.

Construction and testing

An actual-size, single-side PCB for the vibration-activated smart CRO probe is shown in Fig. 4 and its component layout in Fig. 5. After assembling the circuit on the PCB, enclose it in a suitable box. Connect 230V AC input across CON1 and CON3. Connect the CRO across CON4.

Fix the vibration sensor on top of the CRO probe as shown in Fig. 1(c). Use a highly-flexible and lightweight wire (like an earphone

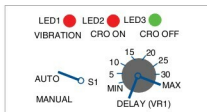


Fig. 6: Proposed front panel

PARTS LIST

Semiconductors:

- IC1 - LM324 op-amp
- T1 - BC547 npn transistor
- D1 - 1N4148 signal diode
- D2 - 1N4007 rectifier diode
- BR1 - DB107 bridge rectifier
- LED1-LED3 - 5mm LED

Resistors (all 1/4-watt, $\pm 5\%$ carbon):

- R1, R2, R7 - 10-kilo-ohm
- R3 - 100-kilo-ohm
- R4, R5, R8 - 2.2-kilo-ohm
- R6 - 15-kilo-ohm
- R9 - 10-mega-ohm
- VR1 - 10-kilo-ohm potmeter

Capacitors:

- C1 - 1000 μ F, 35V electrolytic
- C2 - 100 μ F, 25V electrolytic
- C3 - 1 μ F, 25V electrolytic

Miscellaneous:

- S1 - SPDT switch
- PZ1 - Piezoelectric buzzer
- CON1 - 2-pin AC power plug
- CON2 - 2-pin connector
- CON3-CON4 - 2-pin terminal connector
- X1 - 230V AC primary to 12V, 500mA secondary transformer
- RL1 - 12V, 1C/O relay
- VR1 - HXD-2NC vibration-sensor module
- CRO with probe

wire) to connect the sensor. Connect AC mains supply's live wire L and neutral N across CON3.

Potmeter VR1 is calibrated in terms of time. 10.45V is the maximum C2 charging voltage when sensor is subjected to vibration. During calibration mark the minimum (one minute) and maximum (25 minutes) delays. Proposed front-panel dial is shown in Fig. 6.

Provide proper insulation between AC and DC voltages. Ensure all test points have voltages as per the table before using the circuit. Main DC voltage at TP1 is unregulated. **EFY**



K. Murali Krishna is a former faculty of Aditya Engineering College, Surampalem. Presently, he is working as a telecom technical assistant, BSNL Rajahmundry, Andhra Pradesh.